1 V^S

Given a vector space V, we can form many other vector spaces as follows.

Definition 1.1 (V^S)**.** Let V be a vector space and S a set. Define V^S to be the set of all functions from $S \to V$:

$$V^S := \{f : S \to V\}.$$

V^S is made into a vector space by defining addition and scalar multiplication pointwise:

$$(f+g)(s) := f(s)+g(s)$$
$$(\alpha f)(s) := \alpha f(s).$$

To check the vector space axioms, first recall that two functions $f, g : S \to V$ are equal if and only if $f(s) = g(s)$ for all $s \in S$. Then use the fact that for each $s \in S$ and $f \in V^S$, $f(s) \in V$, and we already know that the vector space axioms hold in V. For example:

(A1): For any $f, g \in V^S$ and $s \in S$, we have

$$\begin{aligned}(f+g)(s) &= f(s)+g(s) && \text{(definition of addition on } V^S\text{)}\\ &= g(s)+f(s) && \text{((A1) in } V\text{)}\\ &= (g+f)(s) && \text{(definition of addition on } V^S\text{)}\end{aligned}$$

which shows that $f+g = g+f$ for all $f, g \in V^S$, hence (A1) holds for addition in V^S.

1. Verify the remaining axioms to prove that V^S is a vector space. Justify each step.

Note that many of the vector spaces we have encountered so far can be viewed as special cases of V^S.

Example 1.2 (F^n)**.** By taking $V = F$ and $S = \{1,2,\ldots,n\}$, V^S is the set of all functions $f : \{1,2,\ldots,n\} \to F$. Such a function is precisely a choice of an n-tuple of elements of F: $(f(1), f(2), \ldots, f(n)) =: (f_1, f_2, \ldots, f_n)$. The set of all ordered n-tuples of elements of F is F^n. The rules for addition and scalar multiplication in V^S become

$$(f+g)(i) = f(i) + g(i) = f_i + g_i$$
$$(\alpha f)(i) = \alpha f(i) = \alpha f_i;$$

that is, we add the corresponding components, and multiply all components by α, which are exactly the definitions of vector addition and scalar multiplication in F^n.

2. Explain how each of the following vector spaces can be realized as F^S for some choice of set S. Don't forget to check that vector addition and scalar multiplication work out as they are supposed to.

   (a) F^ω (the vector space of all infinite sequences of elements of F)

   (b) $F^{m,n}$ (the vector space of all $m \times n$ matrices)

   (c) $\{0\}$ (the zero vector space)